

TANISHKA VATSA

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LinkedIn | GitHub | LeetCode

SUMMARY

B.Tech Computer Science student with hands-on experience in Python, Java, cloud computing, and cybersecurity fundamentals. Experienced in developing security-focused applications using Flask and Python. Proficient in AWS cloud services, Linux, Git, and Data Structures & Algorithms. Passionate about software development, cloud technologies, and cybersecurity, seeking internship opportunities to apply technical skills in real-world environments.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++

Core Subjects: Data Structures & Algorithms, Object-Oriented Programming, Cloud Computing, Cybersecurity Fundamentals

Tools & Technologies: AWS, Git, GitHub, Linux, Flask, Regex

Concepts: Authentication, Password Security, Input Validation, DevOps Fundamentals, NLP Basics

PROJECTS

Secure Login System | Python, Flask, Authentication, Cybersecurity

GitHub: <https://github.com/Tanishka267/secure-login-system/tree/main>

- Developed a secure login system to address authentication vulnerabilities such as weak passwords and brute-force attacks
- Implemented password hashing using SHA-256 and regex-based password strength validation
- Designed account lock mechanism to restrict repeated failed login attempts
- Applied backend security practices including input validation and secure credential handling to build scalable and maintainable authentication workflows

Python Password Strength Checker | Python, Regex

GitHub: <https://github.com/Tanishka267/python-password-strength-checker>

- Built a Python-based application to evaluate password strength using rule-based validation
- Implemented regex-based checks for length, character variety, and special symbols
- Classified passwords into security levels (Weak, Medium, Strong) based on defined criteria
- Applied cybersecurity principles to improve input validation and password security awareness

EXPERIENCE

AI & Cybersecurity Intern | Centre for Development of Advanced Computing (C-DAC), Mohali
June 2026 – Present (45-Day Internship)

- Undergoing hands-on training in Artificial Intelligence and Cybersecurity concepts.
- Working with cybersecurity fundamentals, threat analysis, and security best practices.
- Exploring AI applications, machine learning concepts, and real-world use cases.
- Gaining practical experience with Linux environments, networking concepts, and security tools.
- Participating in project-based learning and technical workshops.

VIRTUAL EXPERIENCE PROGRAMS

Technology Job Simulation – Deloitte (Forage) | Jan 2026

- Completed consulting-based simulation involving real-world business scenarios
- Analyzed system design and cybersecurity risks

Cybersecurity Analyst Simulation – Tata (Forage) | Jan 2026

- Assessed cybersecurity risks
- Explored security frameworks

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science Engineering

Amity University Punjab | Expected Graduation: 2028

CGPA: 7.32/10

Relevant Coursework: Data Structures & Algorithms, Python Programming, Cloud Computing, Cybersecurity

ACHIEVEMENTS

- Solved 25+ Data Structures & Algorithms problems on LeetCode.
- Completed AWS Academy Cloud Foundations and Cloud Developing certifications.
- Developed cybersecurity-focused projects demonstrating authentication and password security concepts.

CERTIFICATIONS

- AWS Academy Graduate – Cloud Developing
- AWS Academy Graduate – Cloud Foundations
- Introduction to Cybersecurity – Cisco
- The Joy of Computing Using Python – NPTEL
- Machine Learning Crash Course: Logistic & Linear Regression – Google
- Getting Started with Generative AI – IBM
- Introduction to Transformer-Based Natural Language Processing – NVIDIA
- Java Programming Fundamentals – Infosys Springboard
- Fundamentals of DevOps – HCL GUVI
- Anthropic Claude 101
- MongoDB Vector Search Performance

